

The Gaussian Correlation Inequality

Slide edition

After T. Royen (2014) and R. Łatała & D. Matlak (2015)

1 Introduction

Theorem 1.1 (Gaussian correlation inequality). For any closed symmetric convex sets $K, L \subseteq \mathbb{R}^d$ and any centered Gaussian measure μ on $\mathbb{R}^d$,

$$\mu(K \cap L) \geq \mu(K) \mu(L). \quad (1)$$

These slides prove Theorem 1.1 by Royen's analytic argument, following Latała–Matlak's Gaussian-only exposition. The inputs are the Laplace transform of a squared Gaussian, a determinant expansion, dominated convergence, and integration by parts.

Notation 1.2.

- $\mathcal{N}(0, C)$: centered Gaussian distribution with covariance C .
- $M_{n \times m}$: real $n \times m$ matrices. $A \succ 0$ (resp. $A \succcurlyeq 0$) means A is symmetric positive (semi)definite.
- For $A = (a_{ij})_{i,j \leq n}$ and $J \subseteq [n] := \{1, \dots, n\}$, $A_J := (a_{ij})_{i,j \in J}$ is the principal submatrix, $|J|$ the cardinality, and $|A| = \sum_{\sigma} \text{sgn}(\sigma) \prod_i a_{i\sigma(i)}$ the determinant (Leibniz formula, σ over the permutations of $[n]$); we use freely $|AB| = |A||B|$, row-operation invariance and the block-triangular rule.
- I_n : $n \times n$ identity.
- For $\varepsilon \in \{-1, 1\}^n$, $x \in (0, \infty)^n$: $\sqrt{x}_\varepsilon := (\varepsilon_i \sqrt{x_i})_{i \leq n}$.

We will use three standard facts without further comment.

Lemma 1.3 (Square root). For any $A \succcurlyeq 0$ there exists a unique $A^{1/2} \succcurlyeq 0$ with $(A^{1/2})^2 = A$, and $A^{1/2}$ is invertible iff $A \succ 0$.

(Spectral theorem: diagonalize $A = O\Lambda O^\top$ with $\Lambda \succcurlyeq 0$, set $A^{1/2} := O\Lambda^{1/2}O^\top$; uniqueness is standard.)

Lemma 1.4 (Sylvester determinant identity). For $A_1 \in M_{n \times m}$ and $A_2 \in M_{m \times n}$, $|I_n + A_1 A_2| = |I_m + A_2 A_1|$.

Proof. Compute $D := \begin{vmatrix} I_n & -A_1 \\ A_2 & I_m \end{vmatrix}$ two ways. Row-reduce $R_1 \leftarrow R_1 + A_1 R_2$ to clear the $(1, 1)$ -block's contribution to the $(1, 2)$ block: the matrix becomes $\begin{pmatrix} I_n + A_1 A_2 & 0 \\ A_2 & I_m \end{pmatrix}$, so $D = |I_n + A_1 A_2|$. Alternatively, column-reduce $C_1 \leftarrow C_1 + C_2 A_2$: the matrix becomes $\begin{pmatrix} I_n & -A_1 \\ 0 & I_m + A_2 A_1 \end{pmatrix}$, so $D = |I_m + A_2 A_1|$. $\square$

Lemma 1.5 (Strict separation; Hahn–Banach). Let $K_1, K_2 \subseteq \mathbb{R}^d$ be disjoint closed convex sets with K_1 bounded. There exist $w \in \mathbb{R}^d \setminus \{0\}$ and $\alpha_1 < \alpha_2$ in $\mathbb{R}$ with $\langle w, x \rangle \leq \alpha_1$ for $x \in K_1$ and $\langle w, x \rangle \geq \alpha_2$ for $x \in K_2$.

(Standard corollary of Hahn–Banach in finite dimension; see e.g. Rockafellar, *Convex Analysis*, Cor. 11.4.2.)

2 Setup and reduction to rectangles

Definition 2.1 (Centered Gaussian measure). For $C \succ 0$, the measure $\mathcal{N}(0, C)$ on $\mathbb{R}^n$ has density

$$\phi_C(x) = (2\pi)^{-n/2} |C|^{-1/2} \exp\left(-\frac{1}{2} \langle C^{-1}x, x \rangle\right).$$

More generally, a random vector X in $\mathbb{R}^n$ is *centered Gaussian with covariance* $C \succcurlyeq 0$ if $X = AZ$ for some $A \in M_{n \times m}$ with $AA^\top = C$ and $Z \sim \mathcal{N}(0, I_m)$. Its law depends only on C (its characteristic function is $u \mapsto e^{-\langle Cu, u \rangle/2}$); we denote it $\mathcal{N}(0, C)$, and for $C \succ 0$ it is the measure with density ϕ_C above. A centered Gaussian measure is a measure $\mathcal{N}(0, C)$ with $C \succcurlyeq 0$.

If $G = AZ \sim \mathcal{N}(0, \Sigma)$ on $\mathbb{R}^d$ and $V \in M_{n \times d}$, then $VG = (VA)Z$ is centered Gaussian with covariance $V\Sigma V^\top$, which may be singular even when $\Sigma \succ 0$ (e.g. whenever $n > d$). We assume $C \succ 0$ throughout the proof of Theorem 2.4; singular $C \succcurlyeq 0$ is reduced to it by Theorem 2.5 below.

Definition 2.2. $K \subseteq \mathbb{R}^n$ is *symmetric* if $K = -K$. A *symmetric slab* is $S_{v,t} := \{x \in \mathbb{R}^n : |\langle v, x \rangle| \leq t\}$ for $v \in \mathbb{R}^n \setminus \{0\}$, $t > 0$.

Lemma 2.3 (Slab approximation). Every nonempty *bounded* closed symmetric convex set $K \subseteq \mathbb{R}^d$ is a countable intersection of symmetric slabs $S_{v,t}$.

Proof. Fix $R < \infty$ with $K \subseteq B(0, R)$, and let $\mathcal{V} := \{(v, t) \in (\mathbb{Q}^d \setminus \{0\}) \times \mathbb{Q}_{>0} : K \subseteq S_{v,t}\}$, countable; trivially $K \subseteq \bigcap_{(v,t) \in \mathcal{V}} S_{v,t}$. For the reverse inclusion, each $y \notin K$ must be excluded by some slab in $\mathcal{V}$.

Fix $y \notin K$ and $\varepsilon > 0$ with $B(y, \varepsilon) \cap K = \emptyset$. Theorem 1.5 applied to $\overline{B(y, \varepsilon/2)}$ (bounded) and K (closed convex), after a sign change if necessary, gives $w \in \mathbb{R}^d \setminus \{0\}$ and $\alpha_1 < \alpha_2$ with $\langle w, x \rangle \leq \alpha_1$ on K and $\langle w, x \rangle \geq \alpha_2$ on $\overline{B(y, \varepsilon/2)}$. Set $a := \sup_{x \in K} \langle w, x \rangle \leq \alpha_1$. Since K is nonempty, symmetric and convex, $0 = \frac{1}{2}x + \frac{1}{2}(-x) \in K$, so $a \geq 0$; and $K = -K$ gives $\sup_{x \in K} |\langle w, x \rangle| \leq a$. The minimum of $\langle w, \cdot \rangle$ on $\overline{B(y, \varepsilon/2)}$ is at $y - \frac{\varepsilon}{2}w/|w|$, so

$$\langle w, y \rangle = \min_{z \in \overline{B(y, \varepsilon/2)}} \langle w, z \rangle + \frac{\varepsilon}{2}|w| \geq a + \frac{\varepsilon}{2}|w| > 0.$$

Now fix $0 < \delta < \frac{\varepsilon|w|}{2(|y| + R + 1)}$ and choose $w' \in \mathbb{Q}^d$ with $|w' - w| < \delta$. Then $w' \neq 0$: since $0 \in K$, $\varepsilon \leq |y|$, so $\delta < |w|/2$ and $|w'| > |w|/2 > 0$. Using $|x| \leq R$ on K ,

$$\sup_{x \in K} |\langle w', x \rangle| \leq a + \delta R, \quad |\langle w', y \rangle| \geq a + \frac{\varepsilon}{2}|w| - \delta|y|.$$

Take a rational $a' \in [a + \delta R, a + \delta R + \delta/2]$ (so $a' > 0$). The choice of δ gives $\delta(|y| + R + \frac{1}{2}) < \frac{\varepsilon}{2}|w|$, hence $K \subseteq S_{w', a'}$ and $y \notin S_{w', a'}$. $\square$

Reduction to slabs. If K or L is empty, both sides of (1) vanish; assume both nonempty. Truncating $K_m := K \cap [-m, m]^d$, $L_m := L \cap [-m, m]^d$ (nonempty since $0 \in K \cap L$; bounded closed symmetric convex; increasing to

K, L), continuity of μ from below reduces the GCI to *bounded* K, L . Then Theorem 2.3 writes each as a countable intersection of symmetric slabs; intersecting the first N and letting $N \rightarrow \infty$, continuity from above reduces the GCI to *finite* intersections of symmetric slabs. Writing

$$K = \{x : |\langle v_i, x \rangle| \leq t_i, 1 \leq i \leq n_1\}, \quad L = \{x : |\langle v_i, x \rangle| \leq t_i, n_1+1 \leq i \leq n\},$$

setting $n = n_1 + n_2$ and $X_i := \langle v_i, G \rangle$ for $G \sim \mu$, the vector $X = (X_1, \dots, X_n)$ is centered Gaussian with a covariance $C \succcurlyeq 0$ that is in general singular, and the GCI reduces to the following theorem together with Theorem 2.5:

Theorem 2.4 (GCI, analytic form). Let $X = (X_1, \dots, X_n)$ be centered Gaussian with $C \succcurlyeq 0$, $n = n_1 + n_2$. For any $t_1, \dots, t_n > 0$,

$$\mathbf{P}(|X_i| \leq t_i, i \leq n) \geq \mathbf{P}(|X_i| \leq t_i, i \leq n_1) \mathbf{P}(|X_i| \leq t_i, n_1+1 \leq i \leq n). \quad (2)$$

Lemma 2.5 (Singular covariance). If (2) holds for every centered Gaussian X with covariance $C \succcurlyeq 0$, then it holds for every centered Gaussian X with covariance $C \succcurlyeq 0$ (and the same $t_i > 0$).

Proof. Let $X = AZ$ have covariance $C \succcurlyeq 0$, let $Y \sim \mathcal{N}(0, I_n)$ be independent of Z , and for $\varepsilon > 0$ put $X^\varepsilon := X + \varepsilon Y = \begin{bmatrix} A & \varepsilon I_n \end{bmatrix} \begin{pmatrix} Z \\ Y \end{pmatrix}$, centered Gaussian with covariance $C + \varepsilon^2 I_n \succcurlyeq 0$; so (2) holds for X^ε . For each i , $\mathbf{P}(|X_i| = t_i) = 0$: if $C_{ii} > 0$ then X_i has a density, and if $C_{ii} = 0$ then $X_i = 0 < t_i$ a.s. As $\varepsilon \rightarrow 0$, $X_i^\varepsilon \rightarrow X_i$ pointwise, so on the full-probability event $\bigcap_i \{|X_i| \neq t_i\}$ every indicator $\mathbf{1}\{|X_i^\varepsilon| \leq t_i\}$, and hence every product of them, converges to its value for X . By dominated convergence (bound 1), the three probabilities in

(2) converge, and the inequality passes to the limit. □

3 Auxiliary lemmas

3.1 Laplace transform of squared Gaussian vectors

Lemma 3.1. For $X \sim \mathcal{N}(0, C)$ in $\mathbb{R}^n$ and $\lambda_i \geq 0$,

$$\mathbb{E} \exp\left(-\sum_{i=1}^n \lambda_i X_i^2\right) = |I_n + 2\Lambda C|^{-1/2}, \quad \Lambda := \text{diag}(\lambda_1, \dots, \lambda_n).$$

Proof.

1. Standard normal quadratic. For $Y \sim \mathcal{N}(0, I_n)$ and a symmetric matrix B with $I_n - 2B \succ 0$, combine the density of Y with the factor $e^{\langle BY, Y \rangle}$: the exponent is $-\frac{1}{2}\langle y, y \rangle + \langle By, y \rangle = -\frac{1}{2}\langle (I_n - 2B)y, y \rangle$, so

$$\mathbb{E} e^{\langle BY, Y \rangle} = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-\frac{1}{2}\langle (I_n - 2B)y, y \rangle} dy = |I_n - 2B|^{-1/2}, \quad (3)$$

using $\int_{\mathbb{R}^n} e^{-\frac{1}{2}\langle Ay, y \rangle} dy = (2\pi)^{n/2} |A|^{-1/2}$ (change variables $y = A^{-1/2}z$; $A = I_n - 2B$).

2. Reduction to the standard case. Set $A := C^{1/2}$ (Theorem 1.3); then $C = AA^\top$ and $X := AY \sim \mathcal{N}(0, C)$. Taking $B := -A^\top \Lambda A$ in (3): since $\lambda_i \geq 0$ gives $\Lambda \succcurlyeq 0$, hence $B \preccurlyeq 0$ and $I_n - 2B \succcurlyeq I_n \succ 0$. The exponent

becomes $\langle BY, Y \rangle = -\langle \Lambda AY, AY \rangle = -\sum_i \lambda_i (AY)_i^2 = -\sum_i \lambda_i X_i^2$, so

$$\mathbb{E} \exp\left(-\sum_i \lambda_i X_i^2\right) = |I_n + 2A^\top \Lambda A|^{-1/2}.$$

3. Commuting the factors. Apply Theorem 1.4 with $A_1 := A^\top$, $A_2 := 2\Lambda A$: $|I_n + 2A^\top \Lambda A| = |I_n + 2\Lambda A A^\top| = |I_n + 2\Lambda C|$.

□

3.2 A determinant identity

Lemma 3.2.

- (i) For $A \in M_{n \times n}$,

$$|I_n + A| = 1 + \sum_{\emptyset \neq J \subseteq [n]} |A_J|. \quad (4)$$

- (ii) For $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$ with A_{ii} invertible,

$$|A| = |A_{11}| |A_{22}| \cdot |I_{n_1} - A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2}|. \quad (5)$$

If $A \succ 0$, then $0 \leq A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2} \leq I_{n_1}$.

Proof.

- (i) By the Leibniz formula for the determinant, $|I_n + A| = \sum_{\sigma} \text{sgn}(\sigma) \prod_i (\delta_{i\sigma(i)} + a_{i\sigma(i)})$. Expanding the product, each summand corresponds to a choice of $J \subseteq [n]$ (positions where $a_{i\sigma(i)}$ is selected); non-zero contributions require $\sigma(i) = i$ for $i \notin J$. The σ 's fixing $[n] \setminus J$ correspond to permutations of J , giving $\sum_J |A_J|$; the $J = \emptyset$ term is 1.
- (ii) Let $A_{11}^{1/2}, A_{22}^{1/2}$ be the positive symmetric square roots (Theorem 1.3), invertible by assumption. Setting $P := A_{11}^{-1/2} A_{12} A_{22}^{-1/2}$ and $Q := A_{22}^{-1/2} A_{21} A_{11}^{-1/2}$, a direct block multiplication gives

$$A = \text{diag}(A_{11}^{1/2}, A_{22}^{1/2}) \begin{pmatrix} I & P \\ Q & I \end{pmatrix} \text{diag}(A_{11}^{1/2}, A_{22}^{1/2}),$$

so $|A| = |A_{11}| |A_{22}| \cdot \left| \begin{pmatrix} I & P \\ Q & I \end{pmatrix} \right|$. Row-reduce by $R_1 \leftarrow R_1 - PR_2$ (determinant invariant):

$$\begin{pmatrix} I & P \\ Q & I \end{pmatrix} \rightarrow \begin{pmatrix} I - PQ & 0 \\ Q & I \end{pmatrix},$$

block-triangular with determinant $|I - PQ|$. Substituting $PQ = A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2}$ gives (5).

When $A \succ 0$: for all $s, t \in \mathbb{R}$ and $x \in \mathbb{R}^{n_1}, y \in \mathbb{R}^{n_2}$,

$$\langle A(sx, ty)^\top, (sx, ty)^\top \rangle = s^2 \langle A_{11}x, x \rangle + 2st \langle A_{21}x, y \rangle + t^2 \langle A_{22}y, y \rangle \geq 0.$$

Nonnegativity of this quadratic in (s, t) forces $\langle A_{21}x, y \rangle^2 \leq \langle A_{11}x, x \rangle \langle A_{22}y, y \rangle$. Setting $x := A_{11}^{-1/2}u$, $y := A_{22}^{-1/2}v$: $\langle Qu, v \rangle^2 \leq |u|^2 |v|^2$, i.e., $\|Q\|_{\text{op}} \leq 1$, equivalently $0 \preceq Q^\top Q \preceq I_{n_1}$. For A symmetric, $Q = P^\top$, so $A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2} = PP^\top = Q^\top Q$.

□

3.3 Density of $Z(\tau)$ and the interchange lemma

Partition

$$C = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}, \quad C_{ij} \in M_{n_i \times n_j}, \quad n_1 + n_2 = n, \quad (6)$$

and define the interpolating family

$$C(\tau) := \begin{pmatrix} C_{11} & \tau C_{12} \\ \tau C_{21} & C_{22} \end{pmatrix} = \tau C(1) + (1 - \tau)C(0), \quad \tau \in [0, 1]. \quad (7)$$

Since $C(0) = \text{diag}(C_{11}, C_{22}) \succ 0$ and $C(1) = C \succ 0$, each $C(\tau) \succ 0$ by convexity. Let $X(\tau) \sim \mathcal{N}(0, C(\tau))$ and $Z_i(\tau) := \frac{1}{2}X_i(\tau)^2$.

Lemma 3.3 (Density and interchange). Let $f(x, \tau)$ denote the density of $Z(\tau)$ on $(0, \infty)^n$.

(i)

$$f(x, \tau) = |C(\tau)|^{-1/2} (4\pi)^{-n/2} \frac{1}{\sqrt{x_1 \cdots x_n}} \sum_{\varepsilon \in \{-1, 1\}^n} e^{-\langle C(\tau)^{-1} \sqrt{x_\varepsilon}, \sqrt{x_\varepsilon} \rangle} \mathbf{1}_{(0, \infty)^n}(x). \quad (8)$$

(ii) For any Borel $K \subseteq [0, \infty)^n$ and $\lambda_i \geq 0$,

$$\int_K e^{-\sum_i \lambda_i x_i} \frac{\partial}{\partial \tau} f(x, \tau) dx = \frac{\partial}{\partial \tau} \int_K e^{-\sum_i \lambda_i x_i} f(x, \tau) dx. \quad (9)$$

Proof.

- (i) The density of $X(\tau)$ is $\phi_{C(\tau)}$ from Theorem 2.1. On each sign orthant $\mathbb{R}_\varepsilon^n := \{y : \text{sgn}(y_i) = \varepsilon_i\}$, the map $y \mapsto x_i := \frac{1}{2}y_i^2$ is a diffeomorphism onto $(0, \infty)^n$ with inverse $y_i = \varepsilon_i\sqrt{2x_i}$ and Jacobian $dy_i/dx_i = \varepsilon_i/\sqrt{2x_i}$; absolute Jacobian $\prod_i(2x_i)^{-1/2}$. Summing over the 2^n sign patterns and using $\langle C(\tau)^{-1}y, y \rangle = 2\langle C(\tau)^{-1}\sqrt{x_\varepsilon}, \sqrt{x_\varepsilon} \rangle$ (by $y_i = \varepsilon_i\sqrt{2x_i}$):

$$\begin{aligned} f(x, \tau) &= \sum_{\varepsilon} \phi_{C(\tau)}(\sqrt{2x_\varepsilon}) \cdot \prod_i (2x_i)^{-1/2} \\ &= |C(\tau)|^{-1/2} (2\pi)^{-n/2} 2^{-n/2} (x_1 \cdots x_n)^{-1/2} \sum_{\varepsilon} e^{-\langle C(\tau)^{-1}\sqrt{x_\varepsilon}, \sqrt{x_\varepsilon} \rangle}, \end{aligned}$$

which is (8) after combining $(2\pi)^{-n/2} 2^{-n/2} = (4\pi)^{-n/2}$.

- (ii) We build a τ -uniform integrable dominating function for $\partial_\tau f$ via four steps.
- (a) Eigenvalue lower bound. Each $C(\tau) \succ 0$, and $\tau \mapsto \lambda_{\min}(C(\tau)^{-1}) = 1/\lambda_{\max}(C(\tau))$ is continuous and positive on compact $[0, 1]$, so bounded below by some $a > 0$. Then $\langle C(\tau)^{-1}\sqrt{x_\varepsilon}, \sqrt{x_\varepsilon} \rangle \geq a|\sqrt{x_\varepsilon}|^2 = a \sum_i x_i$.
- (b) Derivative formula. Differentiating $C(\tau)C(\tau)^{-1} = I_n$, $\frac{d}{d\tau}C(\tau)^{-1} = -C(\tau)^{-1}C'(\tau)C(\tau)^{-1}$, with $C'(\tau) = C(1) - C(0)$ constant. Since $\tau \mapsto C(\tau)^{-1}$ is continuous on compact $[0, 1]$, $\|\frac{d}{d\tau}C(\tau)^{-1}\|_{\text{op}}$ is bounded, say by $b < \infty$. Since $|C(\tau)|$ is a polynomial in τ , positive on $[0, 1]$, the map $\tau \mapsto |C(\tau)|^{-1/2}$ is C^1 on $[0, 1]$, so it and its derivative are bounded there, say by $M, M' < \infty$. Hence $|C(\tau)|^{-1/2} \leq M$, $|\partial_\tau |C(\tau)|^{-1/2}| \leq M'$, and $|\langle \frac{d}{d\tau}C(\tau)^{-1}\sqrt{x_\varepsilon}, \sqrt{x_\varepsilon} \rangle| \leq b|\sqrt{x_\varepsilon}|^2 = b \sum_i x_i$.

- (c) Dominating function. Differentiating (8) in τ (product rule on the prefactor $|C(\tau)|^{-1/2}$ and the exponent), each of the 2^n sign-orthant summands is bounded by

$$(M' + Mb \sum_i x_i)(4\pi)^{-n/2}(x_1 \cdots x_n)^{-1/2} e^{-a \sum_i x_i}.$$

Summing over ε and absorbing constants,

$$\sup_{\tau \in [0,1]} |\partial_\tau f(x, \tau)| \leq g(x) := c \frac{1 + \sum_i x_i}{\sqrt{x_1 \cdots x_n}} e^{-a \sum_i x_i}$$

for some $c = c(C, n)$. Since $1 + \sum_i x_i \leq \prod_i (1 + x_i)$, $g(x) \leq c \prod_i x_i^{-1/2} (1 + x_i) e^{-ax_i}$, which factorises; as $\int_0^\infty x^{-1/2} (1 + x) e^{-ax} dx < \infty$, Tonelli gives $\int g < \infty$.

- (d) Interchange. For any Borel $K \subseteq [0, \infty)^n$ and $\lambda_i \geq 0$, $e^{-\sum_i \lambda_i x_i} |\partial_\tau f| \leq g$ uniformly in τ . Dominated convergence applied to the difference quotient $(f(x, \tau + h) - f(x, \tau))/h \rightarrow \partial_\tau f(x, \tau)$ gives (9). □

3.4 The noncentral gamma and the h -family

Definition 3.4. For $\alpha > 0$, $y \geq 0$, the *noncentral gamma density* is

$$g_\alpha(x, y) := e^{-x-y} \sum_{k=0}^{\infty} \frac{x^{k+\alpha-1}}{\Gamma(k+\alpha)} \frac{y^k}{k!}, \quad x > 0.$$

Remark 3.5. Equivalently,

$$g_\alpha(x, y) = \sum_{k \geq 0} e^{-y} \frac{y^k}{k!} g_{\alpha+k}(x),$$

a Poisson(y) mixture of central gamma($\alpha + k$) densities; so $g_\alpha(\cdot, y)$ is a probability density for each $y \geq 0$.

Definition 3.6. For $\mu > 0$, $\alpha \in (0, \infty)^n$, and $Y = (Y_1, \dots, Y_n)$ with $Y_i \geq 0$ a.s.,

$$h_{\alpha, \mu, Y}(x) := \mathbb{E} \prod_{i=1}^n \frac{1}{\mu} g_{\alpha_i} \left(\frac{x_i}{\mu}, Y_i \right), \quad x \in (0, \infty)^n.$$

Lemma 3.7 (Properties of the h -family). Write $h_\alpha := h_{\alpha,\mu,Y}$.

- (i) $h_\alpha \geq 0$ and $\int h_\alpha dx = 1$.
- (ii) If $\alpha_i > 1$, then $h_\alpha(x) \rightarrow 0$ as $x_i \rightarrow 0+$, and

$$\frac{\partial}{\partial x_i} h_\alpha = \frac{1}{\mu} (h_{\alpha - e_i} - h_\alpha). \quad (10)$$

- (iii) If $\alpha \in (1, \infty)^n$, then for any $J \subseteq [n]$ and $\lambda_i \geq 0$, $\partial^{|J|} h_\alpha / \partial x_J \in L_1((0, \infty)^n)$ and

$$\int e^{-\sum_i \lambda_i x_i} \frac{\partial^{|J|}}{\partial x_J} h_\alpha dx = \left(\prod_{i \in J} \lambda_i \right) \int e^{-\sum_i \lambda_i x_i} h_\alpha dx. \quad (11)$$

Proof.

- (i) $\int_0^\infty \mu^{-1} g_\alpha(x_i/\mu, Y_i) dx_i = \int_0^\infty g_\alpha(u, Y_i) du = 1$ for each ω (Theorem 3.5); Fubini gives $\int h_\alpha = \mathbb{E}[1] = 1$.

- (ii) We split into two parts.

- (a) Gamma bound. $\log \Gamma$ is strictly convex on $(0, \infty)$, and since $\Gamma(1) = \Gamma(2) = 1$, Γ has a unique minimum at some $s_0 \in (1, 2)$ with $\Gamma(s_0) > \frac{1}{2}$ (numerically $s_0 \approx 1.462$, $\Gamma(s_0) \approx 0.886$). Hence

$$\Gamma(s) \geq \frac{1}{2} \quad \text{for all } s > 0, \quad (12)$$

giving $\Gamma(\alpha)^{-1} \leq 2$ for the $k = 0$ term. For $k \geq 1$, induction via $\Gamma(k + \alpha) = (k - 1 + \alpha)\Gamma(k - 1 + \alpha) \geq k \cdot \frac{1}{2}(k - 2)! = \frac{1}{2}(k - 1)!$ gives $\Gamma(k + \alpha) \geq \frac{1}{2}(k - 1)!$ (base case $k = 1$: $\Gamma(1 + \alpha) \geq \frac{1}{2}$). Substituting

into $g_\alpha(x, y)$:

$$\begin{aligned} g_\alpha(x, y) &\leq e^{-x-y} \left[2x^{\alpha-1} + \sum_{k \geq 1} \frac{2x^{k+\alpha-1}y^k}{(k-1)!k!} \right] \\ &= 2x^{\alpha-1}e^{-x-y} + 2x^\alpha e^{-x} \underbrace{\sum_{k \geq 1} \frac{x^{k-1}}{(k-1)!} \cdot \frac{y^k e^{-y}}{k!}}_{\leq e^x \cdot 1} \leq 2x^{\alpha-1}(e^{-x} + x), \end{aligned}$$

using $y^k e^{-y}/k! \leq \sup_y (y^k e^{-y}/k!) \leq 1$ (term in a Poisson distribution), and $\sum_{k \geq 1} x^{k-1}/(k-1)! = e^x$. Thus

$$g_\alpha(x, y) \leq 2x^{\alpha-1}(e^{-x} + x), \quad (13)$$

and substituting into Theorem 3.6 (note Y_i drops out),

$$h_\alpha(x) \leq (2/\mu)^n \prod_i (x_i/\mu)^{\alpha_i-1} (1 + x_i/\mu). \quad (14)$$

When $\alpha_i > 1$, the factor $(x_i/\mu)^{\alpha_i-1} \rightarrow 0$ as $x_i \rightarrow 0+$, so $h_\alpha(x) \rightarrow 0$.

- (b) Derivative formula. Differentiating the series in Theorem 3.4 term by term (using $\Gamma(k + \alpha) = (k + \alpha - 1)\Gamma(k + \alpha - 1)$, valid when $\alpha > 0$),

$$\partial_x g_\alpha(x, y) = g_{\alpha-1}(x, y) - g_\alpha(x, y), \quad \alpha > 1, \quad (15)$$

with convergence locally uniform by (13) applied to $g_{\alpha-1}$ (valid since $\alpha - 1 > 0$). Inserting into Theorem 3.6: the integrand and its x_i -derivative are bounded by an integrable function of (x, ω) uniformly in x_i on compact subsets of $(0, \infty)$ (use (13) for $g_{\alpha_i-1}(\cdot, Y_i)$ and $g_{\alpha_i}(\cdot, Y_i)$), justifying dominated convergence to differentiate under $\mathbf{E}$, which gives (10).

(iii) Iterating (10),

$$\frac{\partial^{|J|}}{\partial x_J} h_\alpha = \frac{1}{\mu^{|J|}} \sum_{\delta \in \{0,1\}^J} (-1)^{|J| - |\delta|} h_{\alpha - \sum_{i \in J} \delta_i e_i}, \quad (16)$$

each summand in L_1 by (i) (all $\alpha_i - \delta_i \geq \alpha_i - 1 > 0$). Integration by parts in each x_j , $j \in J$: the boundary term at $x_j \rightarrow \infty$ vanishes by (14) (exponential decay in every coordinate via $\int_0^\infty x^{\alpha-1} (1+x) e^{-ax} dx < \infty$, implying pointwise decay of the partial integral); at $x_j \rightarrow 0+$ by (ii), applied to the $\alpha'_j := \alpha_j - \delta_j \geq \alpha_j - 1 > 0$ shape, together with the extra factor $e^{-\sum_i \lambda_i x_i}$ which is bounded. Each integration by parts brings down a λ_j , giving (11).

□

3.5 The family $h_{k,C}$ and its Laplace transform

Definition 3.8. For $C \succ 0$ and $\mu > 0$ with $C - \mu I_n \succ 0$, set $A := (C - \mu I_n)^{1/2}$ (Theorem 1.3, which is well-defined and invertible), so that $C = \mu I_n + AA^\top$. Let $(\xi_j^{(l)})$, $j \leq n$, $l \leq k$, be i.i.d. $\mathcal{N}(0, 1)$, and

$$Y_i := \sum_{l=1}^k \left(\sum_{j=1}^n \frac{1}{\sqrt{2\mu}} \xi_j^{(l)} a_{ij} \right)^2, \quad h_{k,C} := h_{(k/2, \dots, k/2), \mu, Y}. \quad (17)$$

Lemma 3.9. For $\lambda_i \geq 0$, $\int e^{-\sum_i \lambda_i x_i} h_{k,C} dx = |I_n + \Lambda C|^{-k/2}$.

Proof.

1. Scalar Laplace transform. Substitute $u := x/\mu$ and expand the series in Theorem 3.4:

$$\begin{aligned} \int_0^\infty \mu^{-1} e^{-\lambda x} g_\alpha(x/\mu, y) dx &= \int_0^\infty e^{-\lambda \mu u - u - y} \sum_{k \geq 0} \frac{u^{k+\alpha-1}}{\Gamma(k+\alpha)} \frac{y^k}{k!} du \\ &= e^{-y} \sum_{k \geq 0} \frac{y^k}{k!} (1 + \mu \lambda)^{-(k+\alpha)} \\ &= (1 + \mu \lambda)^{-\alpha} \exp\left(-\frac{\mu \lambda}{1 + \mu \lambda} y\right), \end{aligned}$$

using $\int_0^\infty u^{k+\alpha-1} e^{-(1+\mu\lambda)u} du = \Gamma(k+\alpha)(1+\mu\lambda)^{-(k+\alpha)}$ (Fubini-term-by-term justified by absolute conver-

gence). So

$$\int_0^\infty \mu^{-1} e^{-\lambda x} g_\alpha(x/\mu, y) dx = (1 + \mu\lambda)^{-\alpha} \exp\left(-\frac{\mu\lambda}{1+\mu\lambda}y\right). \quad (18)$$

2. Product form. Applying (18) coordinatewise with $\alpha_i = k/2$ and Fubini (nonneg integrand),

$$\int e^{-\sum_i \lambda_i x_i} h_{k,C} dx = \prod_i (1 + \mu\lambda_i)^{-k/2} \cdot \mathbb{E} \exp\left(-\sum_i \frac{\mu\lambda_i}{1+\mu\lambda_i} Y_i\right). \quad (19)$$

3. Evaluate the expectation. From (17), set $X^{(l)} := \frac{1}{\sqrt{2\mu}} A \xi^{(l)}$, so $X^{(l)} \sim \mathcal{N}(0, \frac{1}{2\mu} A A^\top)$ i.i.d. across l and $Y_i = \sum_{l=1}^k (X_i^{(l)})^2$. By independence across l and Theorem 3.1 applied to each $X^{(l)}$ with $\tilde{\Lambda} := \text{diag}(\frac{\mu\lambda_i}{1+\mu\lambda_i})$,

$$\mathbb{E} \exp\left(-\sum_i \frac{\mu\lambda_i}{1+\mu\lambda_i} Y_i\right) = \left(\mathbb{E} \exp(-\langle \tilde{\Lambda} X^{(1)}, X^{(1)} \rangle)\right)^k = |I_n + \frac{1}{\mu} \tilde{\Lambda} A A^\top|^{-k/2}.$$

4. Combine. Inserting step 3 into (19) and using $\prod_i (1 + \mu\lambda_i)^{-k/2} = |I_n + \mu\Lambda|^{-k/2}$ (since $I_n + \mu\Lambda$ is diagonal),

$$\int e^{-\sum_i \lambda_i x_i} h_{k,C} dx = (|I_n + \mu\Lambda| \cdot |I_n + \frac{1}{\mu} \tilde{\Lambda} A A^\top|)^{-k/2}.$$

The diagonal identity $\tilde{\Lambda}_i = \mu\lambda_i/(1+\mu\lambda_i)$ rearranges to $(I_n + \mu\Lambda)\tilde{\Lambda} = \mu\Lambda$. Hence, using the multiplicativity

of det,

$$\begin{aligned}
 |I_n + \mu\Lambda| \cdot |I_n + \frac{1}{\mu}\tilde{\Lambda}AA^\top| &= |(I_n + \mu\Lambda)(I_n + \frac{1}{\mu}\tilde{\Lambda}AA^\top)| \\
 &= |I_n + \mu\Lambda + \frac{1}{\mu}(I_n + \mu\Lambda)\tilde{\Lambda}AA^\top| \\
 &= |I_n + \mu\Lambda + \Lambda AA^\top| = |I_n + \Lambda(\mu I_n + AA^\top)| = |I_n + \Lambda C|,
 \end{aligned}$$

giving the claim. □

3.6 Uniqueness of the Laplace transform

Theorem 3.10. If $\varphi_1, \varphi_2 \in L_1((0, \infty)^n)$ satisfy $\int e^{-\sum_i \lambda_i x_i} \varphi_1 dx = \int e^{-\sum_i \lambda_i x_i} \varphi_2 dx$ for all $\lambda \in [0, \infty)^n$, then $\varphi_1 = \varphi_2$ a.e.

Proof. Set $\varphi := \varphi_1 - \varphi_2 \in L_1((0, \infty)^n)$; the hypothesis becomes $\int e^{-\sum_i \lambda_i x_i} \varphi dx = 0$ for all $\lambda \in [0, \infty)^n$.

1. Change of variables. With $y_i := e^{-x_i}$, the map $(0, \infty)^n \rightarrow (0, 1]^n$ has Jacobian $\prod_i y_i^{-1}$, so setting $\tilde{\varphi}(y) := \varphi(-\log y) \prod_i y_i^{-1} \in L_1((0, 1]^n)$ and noting $e^{-\lambda_i x_i} = y_i^{\lambda_i}$,

$$\int_{(0,1]^n} y^\lambda \tilde{\varphi}(y) dy = 0 \quad \text{for all } \lambda \in [0, \infty)^n. \tag{20}$$

2. From polynomials to continuous functions. Restricting (20) to $\lambda \in \mathbb{Z}_{\geq 0}^n$ and taking finite linear combina-

tions, $\int_{[0,1]^n} P(y)\tilde{\varphi}(y) dy = 0$ for every polynomial P (extend $\tilde{\varphi}$ by zero at $y_i = 0$). By Stone–Weierstrass, polynomials are uniformly dense in $C([0,1]^n)$; given $\psi \in C([0,1]^n)$ and $\varepsilon > 0$, pick P with $\|\psi - P\|_\infty < \varepsilon$, so

$$\left| \int \psi \tilde{\varphi} \right| = \left| \int (\psi - P) \tilde{\varphi} \right| \leq \varepsilon \|\tilde{\varphi}\|_1.$$

Sending $\varepsilon \rightarrow 0$: $\int_{[0,1]^n} \psi \tilde{\varphi} dy = 0$ for every $\psi \in C([0,1]^n)$.

3. From continuous functions to $\tilde{\varphi} = 0$ a.e. Let $E := \{\tilde{\varphi} > 0\}$. By inner/outer regularity of Lebesgue measure, there exist compact $K_m \subseteq E$ and open $U_m \supseteq E$ with $|U_m \setminus K_m| \rightarrow 0$; Urysohn gives $\psi_m \in C([0,1]^n)$ with $\mathbf{1}_{K_m} \leq \psi_m \leq \mathbf{1}_{U_m}$. Then $\psi_m \rightarrow \mathbf{1}_E$ a.e. along a subsequence, and $|\psi_m \tilde{\varphi}| \leq |\tilde{\varphi}| \in L_1$, so dominated convergence gives $\int_E \tilde{\varphi} = \lim \int \psi_m \tilde{\varphi} = 0$. Similarly for $\{\tilde{\varphi} < 0\}$, so $\int |\tilde{\varphi}| = 0$ and $\tilde{\varphi} = 0$ a.e., hence $\varphi = 0$ a.e. □

4 Proof of Theorem 2.4

4.1 Reduction to monotonicity

With $s_i := \frac{1}{2}t_i^2$, $K := [0, s_1] \times \cdots \times [0, s_n]$, and $Z(\tau)$ as above, (2) is equivalent to $\mathbf{P}(Z(1) \in K) \geq \mathbf{P}(Z(0) \in K)$. At $\tau = 0$, $C(0) = \text{diag}(C_{11}, C_{22})$ is block-diagonal, so the two blocks are independent and the RHS factorizes. At $\tau = 1$ we recover C . Hence Theorem 2.4 follows from:

Proposition 4.1. $\tau \mapsto \int_K f(x, \tau) dx$ is non-decreasing on $[0, 1]$.

4.2 Laplace-transform computation of $\partial_\tau f$

Theorem 4.1 amounts to $\int_K \partial_\tau f(x, \tau) dx \geq 0$ for each $\tau \in (0, 1)$ (Theorem 3.3(ii) with $\lambda = 0$ justifies $\partial_\tau \int_K f = \int_K \partial_\tau f$). Strategy:

- (a) compute the Laplace transform of $\partial_\tau f$ and match it with that of a sum of derivatives of h_τ ;
- (b) use Laplace injectivity (Theorem 3.10) to get an a.e. pointwise identity;
- (c) integrate this identity over K .

Step (a). Since $Z_i(\tau) = \frac{1}{2}X_i(\tau)^2$, applying Theorem 3.1 to $X(\tau) \sim \mathcal{N}(0, C(\tau))$ with $\lambda_i/2$ in place of λ_i ,

$$\int_{[0, \infty)^n} e^{-\sum_i \lambda_i x_i} f(x, \tau) dx = \mathbb{E} e^{-\sum_i \lambda_i Z_i(\tau)} = |I_n + \Lambda C(\tau)|^{-1/2}.$$

Differentiating in τ , with Theorem 3.3(ii) (taking $K = [0, \infty)^n$) justifying the interchange:

$$\int_{[0, \infty)^n} e^{-\sum_i \lambda_i x_i} \partial_\tau f(x, \tau) dx = \partial_\tau |I_n + \Lambda C(\tau)|^{-1/2}. \quad (21)$$

Expand via Theorem 3.2(i) applied to $A = \Lambda C(\tau)$:

$$|I_n + \Lambda C(\tau)| = 1 + \sum_{\emptyset \neq J \subseteq [n]} |(\Lambda C(\tau))_J| = 1 + \sum_{\emptyset \neq J \subseteq [n]} |C(\tau)_J| \prod_{j \in J} \lambda_j, \quad (22)$$

using $|(\Lambda C(\tau))_J| = |\Lambda_J| |C(\tau)_J| = \prod_{j \in J} \lambda_j \cdot |C(\tau)_J|$ since Λ is diagonal.

The block structure of $|C(\tau)_J|$. For $J \subseteq [n]$, set $J_1 := [n_1] \cap J$ and $J_2 := J \setminus [n_1]$. If $J_1 = \emptyset$ or $J_2 = \emptyset$, then $C(\tau)_J = C_J$ is τ -independent, since the off-diagonal blocks of $C(\tau)$ (where the τ appears) are not sampled. Otherwise, writing $C(\tau)_J$ in block form with blocks indexed by J_1, J_2 ,

$$C(\tau)_J = \begin{pmatrix} C_{J_1} & \tau C_{J_1 J_2} \\ \tau C_{J_2 J_1} & C_{J_2} \end{pmatrix},$$

and applying Theorem 3.2(ii),

$$|C(\tau)_J| = |C_{J_1}| |C_{J_2}| \prod_{i=1}^{|J_1|} (1 - \tau^2 \mu_{J_1, J_2}(i)), \quad (23)$$

where the $\mu_{J_1, J_2}(i) \in [0, 1]$ are the eigenvalues of $C_{J_1}^{-1/2} C_{J_1 J_2} C_{J_2}^{-1} C_{J_2 J_1} C_{J_1}^{-1/2}$ (bounded by Theorem 3.2(ii) since $C_J \succ 0$). Differentiating the product in τ ,

$$\partial_\tau |C(\tau)_J| = -|C_{J_1}| |C_{J_2}| \sum_{i=1}^{|J_1|} 2\tau \mu_{J_1, J_2}(i) \prod_{l \neq i} (1 - \tau^2 \mu_{J_1, J_2}(l)) \leq 0 \quad \text{on } [0, 1].$$

Define

$$a_J(\tau) := -\partial_\tau |C(\tau)_J| \geq 0 \quad (\tau \in [0, 1]), \quad (24)$$

with $a_J := 0$ if $J_1 = \emptyset$ or $J_2 = \emptyset$. Then

$$\partial_\tau |I_n + \Lambda C(\tau)| = - \sum_{\emptyset \neq J \subseteq [n]} a_J(\tau) \prod_{j \in J} \lambda_j. \quad (25)$$

Apply the chain rule $\partial_\tau(u^{-1/2}) = -\frac{1}{2}u^{-3/2}\partial_\tau u$ to $u = |I_n + \Lambda C(\tau)|$: (21) becomes

$$\int_{[0,\infty)^n} e^{-\sum_i \lambda_i x_i} \partial_\tau f(x, \tau) \, dx = -\frac{1}{2} |I_n + \Lambda C(\tau)|^{-3/2} \partial_\tau |I_n + \Lambda C(\tau)|,$$

and substituting (25) (the two minus signs cancel):

$$\int_{[0,\infty)^n} e^{-\sum_i \lambda_i x_i} \partial_\tau f(x, \tau) \, dx = \sum_{\emptyset \neq J \subseteq [n]} \frac{1}{2} a_J(\tau) |I_n + \Lambda C(\tau)|^{-3/2} \prod_{j \in J} \lambda_j. \quad (26)$$

4.3 Identification via $h_\tau := h_{3,C(\tau)}$

By Theorem 3.9 with $k = 3$ and Theorem 3.7(iii) (valid since $\alpha_i = \frac{3}{2} > 1$),

$$\int e^{-\sum_i \lambda_i x_i} \frac{\partial^{|J|}}{\partial x_J} h_\tau \, dx = \left(\prod_{j \in J} \lambda_j \right) |I_n + \Lambda C(\tau)|^{-3/2}.$$

Comparing with (26): $\partial_\tau f(\cdot, \tau)$ and $\sum_J \frac{1}{2} a_J(\tau) \partial^{|J|} h_\tau / \partial x_J$ are L_1 with the same Laplace transform, so by Theorem 3.10,

$$\partial_\tau f(x, \tau) = \sum_{\emptyset \neq J \subseteq [n]} \frac{1}{2} a_J(\tau) \frac{\partial^{|J|}}{\partial x_J} h_\tau(x) \quad \text{a.e. on } (0, \infty)^n. \quad (27)$$

4.4 Integration over K

Integrate (27) over $K = \prod_j [0, s_j]$. For $J^c := [n] \setminus J$ and $(s_J, x_{J^c})_i = s_i$ ($i \in J$), $= x_i$ ($i \in J^c$), the fundamental theorem of calculus in each x_j , $j \in J$, gives

$$\int_K \frac{\partial^{|J|}}{\partial x_J} h_\tau(x) dx = \int_{\prod_{j \in J^c} [0, s_j]} h_\tau(s_J, x_{J^c}) dx_{J^c} \geq 0, \quad (28)$$

since the lower-endpoint contributions vanish by Theorem 3.7(ii) ($\alpha_j = \frac{3}{2} > 1$), and $h_\tau \geq 0$. Since also $a_J \geq 0$,

$$\int_K \partial_\tau f(x, \tau) dx = \sum_{\emptyset \neq J \subseteq [n]} \frac{1}{2} a_J(\tau) \int_{\prod_{j \in J^c} [0, s_j]} h_\tau(s_J, x_{J^c}) dx_{J^c} \geq 0,$$

which proves Theorem 4.1 and hence Theorem 2.4. □

5 Two corollaries

Corollary 5.1 (Šidák). For $X \sim \mathcal{N}(0, C)$ and $s_i \geq 0$,

$$\mathbf{P}(|X_i| \leq s_i, i = 1, \dots, n) \geq \prod_{i=1}^n \mathbf{P}(|X_i| \leq s_i).$$

Proof. If some $s_i = 0$: either $C_{ii} > 0$, so $\mathbf{P}(|X_i| \leq 0) = 0$ and the right side vanishes, or $C_{ii} = 0$, so $X_i = 0$ a.s. and the i -th constraint can be dropped from both sides. So assume all $s_i > 0$, as Theorem 2.4 requires. Induct on n . Case $n = 1$ is trivial. For the step, take $K_1 := \{|x_1| \leq s_1\}$ and $L_1 := \{|x_i| \leq s_i, 2 \leq i \leq n\}$ (both closed symmetric convex). Theorem 2.4 (applied with $n_1 = 1$, $n_2 = n - 1$, and extended to singular C by Theorem 2.5) gives $\mathbf{P}(X \in K_1 \cap L_1) \geq \mathbf{P}(X \in K_1)\mathbf{P}(X \in L_1)$; apply induction to $\mathbf{P}(X \in L_1)$. $\square$

Corollary 5.2 (Decoupling). For $X \sim \mathcal{N}(0, C)$, $C \succcurlyeq 0$, and closed symmetric convex $K, L \subseteq \mathbb{R}^n$,

$$\mathbf{P}(X \in K \cap L) \geq \mathbf{P}(X \in K) \mathbf{P}(X \in L).$$

Proof. Theorem 1.1 with $\mu := \mathcal{N}(0, C)$, a centered Gaussian measure by Theorem 2.1. $\square$